6. Numerical Determination of Optimal Control

In problems with linear plant dynamics and quadratic performance (linear regulator and linear tracking systems), it was found that it is possible to obtain the optimal control law by numerically integrating a matrix differential equation of the Riccati type. Optimal control laws were also determined for several other simple examples by applying Pontryagin's minimum principle.

In general, however, the variational approach leads to a nonlinear two-point boundary-value problem (TPBVP) that cannot be solved analytically to obtain the optimal control law, or even an optimal open-loop control.

Assuming that the state and contol variables are not constrained by any boundaries, that the final time t_f is fixed, and that $\mathbf{x}(t_f)$ is free.

$$\dot{\mathbf{x}}(t) = \frac{\partial H}{\partial \boldsymbol{\lambda}} \tag{1}$$

$$\dot{\boldsymbol{\lambda}}(t) = -\frac{\partial H}{\partial \mathbf{x}} \tag{2}$$

$$\mathbf{0} = \frac{\partial H}{\partial \mathbf{u}} \tag{3}$$

$$\mathbf{x}(t_0) = \mathbf{x}_0 \tag{4}$$

$$\lambda(\mathrm{t_f}) = \frac{\partial \mathrm{h}}{\partial \mathrm{x}(\mathrm{t_f})} \tag{5}$$

An initial guess is used to obtain the solutions to a problem in which one or more of the five necessary and boundary conditions (1) through (5) is not satisfied. This solution is then used to adjust the initial guess in an attempt to make the next solution come closer to satisfying all of the conditions. If these steps are repeated and the iterative procedure converges, the conditions (1) through (5) will eventually be satisfied.

There are two main categories for numerical method to solve the TPBVP.

1) Direct method

Guess control $u(t)$

$$\dot{\mathbf{x}}(t) = \frac{\partial H}{\partial \boldsymbol{\lambda}}$$

$$\dot{\boldsymbol{\lambda}}(t) = -\frac{\partial H}{\partial \mathbf{x}}$$

Modify control: $\mathbf{u}^{k+1}(t) = \mathbf{u}^{k}(t) + \alpha \mathbf{u}(t)$

2) Indirect method

Guess initial costates $\boldsymbol{\lambda}(t_0)$

$$\dot{\mathbf{x}}(t) = \frac{\partial H}{\partial \boldsymbol{\lambda}}$$

$$\dot{\boldsymbol{\lambda}}(t) = -\frac{\partial H}{\partial \mathbf{x}}$$

$\mathbf{0} = \frac{\partial H}{\partial \mathbf{u}}$ or Minimum principle

Update initial costates $\boldsymbol{\lambda}^{k+1}(t_0) = \boldsymbol{\lambda}^{k}(t_0) + \boldsymbol{\delta\lambda}(t_0)$ by using sensitivity matrix at final time.

< *Function Minimization* >

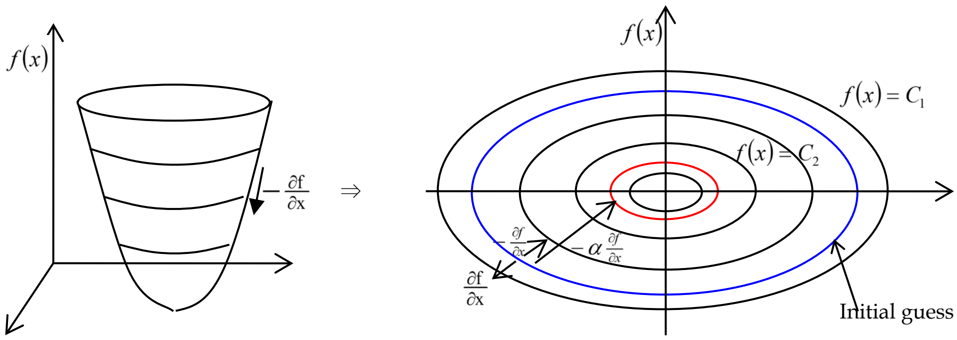
1) 1st order Gradient method (the <u>Steepest Descent</u> Algorithms)

$$Min\ f(x)$$

$$x^{k+1} = x^k - \alpha \frac{\partial f}{\partial x}$$

$\alpha > 0$; step size, α is determined using one-dimension line search

$\frac{\partial f}{\partial x}$; gradient(Jacobian)



If $df(x) = \frac{\partial f}{\partial x}\Delta x = 0$, then f has a minimum

$\left.\frac{\partial f}{\partial x}\right|_{x=x^*} = 0$ at a relative minimum

Convegence is guaranteed if $\alpha \approx 0$ (many iterations)

If function is a particular (so nonlinear) $\rightarrow$ oscillation

2) 2nd order Gradient method (<u>Gauss-Newton</u>)

$$f\left(x^k\right) \approx f\left(x^*\right) + \underbrace{\left.\frac{\partial f}{\partial x}\right|_{x=x^k}}_{G^k;\text{Jacobian}} \Delta x + \frac{1}{2}\Delta x^2 \underbrace{\left.\frac{\partial^2 f}{\partial x^2}\right|_{x=x^k}}_{H^k;\text{Hessian}}$$

x^k ; current guess

$min\ f(x)$ with respect to Δx $\quad (\Rightarrow \frac{\partial f}{\partial \Delta x} = 0)$

$\Rightarrow$ $\min\,[G^k \Delta x + \frac{1}{2}\Delta x^T H^k \Delta x]$

$$\frac{\partial [G^k \Delta x + \frac{1}{2}\Delta x^T H^k \Delta x]}{\partial \Delta x}$$

$$\to G^k + H^k \Delta x = 0$$

$$\to \Delta x = -\left[H^k\right]^{-1} G^k$$

$$\therefore \Delta x^{k+1} = x^k - \alpha\left[H^k\right]^{-1} G^K \qquad \text{step size } \alpha > 0$$

To see the descent property of this method, look at the convection to the formation.

$$\Delta f \approx \frac{\partial f}{\partial x}\Delta x = -G^k\left[H^k\right]^{-1} G^k < 0$$

if H^k is <u>positive definte</u>.

$\Leftrightarrow \Delta f$ is negative

$\to f(x)$ is decreased by each step.

The Hassian has to be positive definite.

Problems 1) for many nonlinear problems, H may not be positive definite.

2) H is expensive to evaluate. → Use <u>Quasi-Newton</u> method

(approximate the inverse of H directly)

< *Simplest Optimal Control Problem (Direct method)* >

$$J = h(\vec{x}(t_f), t_f) + \int_0^{t_f} g(\vec{x}, \vec{u}, t) dt$$

$\vec{x}(0)$ given, $\vec{x}(t_f)$ free, t_f is fixed

$$\delta J_a = \left[\frac{\partial h}{\partial \vec{x}(t_f)} - \vec{\Lambda}^T(t_f)\right]\delta\vec{x}(t_f)$$

$$+\int_0^{t_f}\left[\left(\dot{\vec{\Lambda}}^T + \frac{\partial H}{\partial \vec{x}}\right)^T \delta\vec{x} + \left(\frac{\partial H}{\partial \vec{u}}\right)^T \delta\vec{u} + \left(\frac{\partial H}{\partial \vec{\Lambda}} - \dot{\vec{x}}\right)^T \delta\vec{\Lambda}\right]dt$$

Numerical procedure

1) Guess a time history for $\underline{\vec{u}(t)}$

2) Integrate the state equations forward in time.

$$\dot{\vec{x}} = \vec{f}(\vec{x}, \vec{u}, t)$$

Note down the value of $\vec{x}(t)$

3) Integrate the costate equations backward in time.

$$\dot{\vec{\Lambda}}^T(t_f) = -\frac{\partial H}{\partial \vec{x}} \qquad \text{with} \qquad \vec{\Lambda}^T(t_f) = \frac{\partial h}{\partial \vec{x}(t_f)}$$

Note down the value of $\vec{\lambda}(t)$

4) Note down the value of $\frac{\partial H}{\partial \vec{u}}$ and evaluate $\left\|\frac{\partial H}{\partial \vec{u}}(t)\right\|$.

If $\left\|\frac{\partial H}{\partial \vec{u}}(t)\right\| < \gamma$ (preselected positive constant), terminate the iteration procedure

5) Compute $\delta u^k(t) = -\alpha\frac{\partial H}{\partial u}(t)$ $\rightarrow$ means $\delta u^k(t)$ makes H go to be minimized with respect to *u*. ($\frac{\partial H}{\partial u}(t)$: direction of increasing H by $\delta u^k(t)$),

$$\text{Note}: \ \delta J_a = \int_0^{t_f}\frac{\partial H}{\partial u}(t)\delta u^k(t)dt = \int_0^{t_f}\frac{\partial H}{\partial u}(t)[-\alpha\frac{\partial H}{\partial u}(t)]dt < 0$$

6) Update $\vec{u}(t)^{k+1} = \vec{u}(t)^k + \delta u^k(t)$, $\quad \alpha$; step size

7) Go to step 2)

< *Optimization with terminal constraint specified at t_f* >

Using a Gradient Method

$$J = \phi\left(x\left(t_f\right)\right) + \int_{t_0}^{t_f} L(x,u,t)dt$$

where t_f is specified

$\dot{x} = f(x,u,t), \quad \mathrm{x}\left(\mathrm{t}_0\right) = \mathrm{x}_0$

$\psi\left(x\left(t_f\right)\right) = 0$; terminal constraint

Necessary conditions for a stationary path

Hamiltonian

$$H = L + \Lambda^T f$$

$$\Phi = \phi\left(x\left(t_f\right)\right) + \nu^T \Psi\left(x\left(t_f\right)\right)$$

$$\dot{\Lambda} = -\frac{\partial H}{\partial x} = -\frac{\partial L}{\partial x} - \frac{\partial f}{\partial x}^T \Lambda$$

$$\Lambda\left(t_f\right) = \frac{\partial \Phi}{\partial x\left(t_f\right)} = \left[\frac{\partial \phi}{\partial x} + \nu^T \frac{\partial \psi}{\partial x}\right]_{t=t_f}$$

$$\frac{\partial H}{\partial u} = \frac{\partial L}{\partial u} + \Lambda^T \frac{\partial f}{\partial u} = 0$$

This forms a TPBVP for $x(t)$ and $\Lambda(t)$ where we must also find ν to satisfy the terminal constraints $\psi\left(t_f\right) = 0$

$\psi(x(t_f))$: $q \times 1$ column matrix of terminal conditions

ν : $q \times 1$ column matrix of constants

$\dot{x}$: n first order differential equations of motion

Λ : $n \times 1$ column matrix of costates

u : $m \times 1$ column matrix of controls

A modified (augmented) performance index.

$$J_a = \phi\left(x\left(t_f\right)\right) + \nu^T \psi\left(x\left(t_f\right)\right) + \int_{t_0}^{t_f} \left[L\left(x,u\right) + \Lambda^T \left(f - \dot{x}\right) \right] dt$$

$$\begin{aligned} \delta J_a &= \frac{\partial \phi}{\partial x_f} \delta x_f + \nu^T \frac{\partial \psi}{\partial x_f} \delta x_f \\ &+ \int_{t_0}^{t_f} \left[\left(\frac{\partial L}{\partial x} + \Lambda^T \frac{\partial f}{\partial x} \right) \delta x + \left(\frac{\partial L}{\partial u} + \Lambda^T \frac{\partial f}{\partial u} \right) \delta u - \Lambda^T \delta \dot{x} + \delta \Lambda \left(f - \dot{x}\right) \right] dt \end{aligned}$$

$$\begin{aligned} \delta J_a &= \left[\frac{\partial \phi}{\partial x_f} + \nu^T \frac{\partial \psi}{\partial x_f} - \Lambda\left(t_f\right) \right] \delta x_f \\ &+ \int_{t_0}^{t_f} \left[\left(\frac{\partial L}{\partial x} + \Lambda^T \frac{\partial f}{\partial x} + \dot{\Lambda}^T \right) \delta x + \left(\frac{\partial L}{\partial u} + \Lambda^T \frac{\partial f}{\partial u} \right) \delta u + \delta \Lambda \left(f - \dot{x}\right) \right] dt \end{aligned}$$

$$\begin{aligned} \phi &\rightarrow \Lambda^{\phi} \\ \psi &\rightarrow \Lambda^{\psi} \end{aligned}$$

$$\Phi\left(t_f\right) = \phi\left(x\left(t_f\right)\right) + \nu^T \psi\left(x\left(t_f\right)\right)$$

$$\rightarrow \Lambda = \Lambda^{\phi} + \nu^T \Lambda^{\psi}$$

$$\begin{aligned} \delta J_a &= \left[\frac{\partial \phi}{\partial x_f} + \nu^T \frac{\partial \psi}{\partial x_f} - \Lambda^{\phi}\left(t_f\right) - \nu^T \Lambda^{\psi}\left(t_f\right) \right] \delta x_f \\ &+ \int_{t_0}^{t_f} \left[\begin{array}{l} \left(\frac{\partial L}{\partial x} + \left(\Lambda^{\phi} + \nu^T \Lambda^{\psi}\right) \frac{\partial f}{\partial x} + \left(\dot{\Lambda}^{\phi} + \nu^T \dot{\Lambda}^{\psi}\right)^T \right) \delta x \\ \quad + \left(\frac{\partial L}{\partial u} + \left(\Lambda^{\phi} + \nu^T \Lambda^{\psi}\right)^T \frac{\partial f}{\partial u} \right) \delta u + \delta \Lambda \left(f - \dot{x}\right) \end{array} \right] dt \end{aligned}$$

Let $\Lambda^{\phi}\left(t_f\right) = \frac{\partial \phi}{\partial x_f}$, $\Lambda^{\psi}\left(t_f\right) = \frac{\partial \psi}{\partial x_f}$ $\Rightarrow$ the terms of δx_f goes to zero.

Let $\dot{\Lambda}^{\phi}\left(t\right) = -\frac{\partial L}{\partial x} - \Lambda^{\phi} \frac{\partial f}{\partial x}$, $\dot{\Lambda}^{\psi}\left(t\right) = -\Lambda^{\psi} \frac{\partial f}{\partial x}$ $\Rightarrow$ the terms of δx goes to zero.

Let $\dot{x} = f$ $\Rightarrow$ the terms of $\delta \Lambda$ goes to zero.

$$\Rightarrow \delta J_a = \int_{t_0}^{t_f} \left(\frac{\partial L}{\partial u} + \Lambda^{\phi} \frac{\partial f}{\partial u} + \nu^T \Lambda^{\psi} \frac{\partial f}{\partial u} \right) \delta u \, dt$$

$$\equiv \delta J + \nu^T \delta\psi$$

$$\text{where } \delta J \equiv \int_{t_0}^{t_f} \left(\frac{\partial L}{\partial u} + \Lambda^{\phi} \frac{\partial f}{\partial u} \right) \delta u \, dt, \quad \delta\psi \equiv \int_{t_0}^{t_f} \left(\Lambda^{\psi} \frac{\partial f}{\partial u} \right) \delta u \, dt$$

Let's define $H_u^{\phi} \equiv L_u + \Lambda^{\phi} f_u, \quad H_u^{\psi} \equiv \Lambda^{\psi} f_u$

$$\delta J_a = \int_{t_0}^{t_f} \left(H_u^{\phi} + \nu^T H_u^{\psi} \right) \delta u \, dt$$

For every iteration, δJ_a should be less than zero ($\delta J_a < 0$) →decreasing J

Thus, δu is chosen to be

$$\delta u = -k\left[H_u^{\phi} + \nu^T H_u^{\psi} \right]$$

$$\text{Note: } \delta J_a = \int_{t_0}^{t_f} \left(H_u^{\phi} + \nu^T H_u^{\psi} \right) \delta u \, dt = -k \int_{t_0}^{t_f} \left(H_u^{\phi} + \nu^T H_u^{\psi} \right)^T \left(H_u^{\phi} + \nu^T H_u^{\psi} \right) dt < 0$$

Then,

$$\delta\psi = \int_{t_0}^{t_f} \left(\Lambda^{\Psi} \frac{\partial f}{\partial u} \delta u \right) dt = -k \int_{t_0}^{t_f} \left(\Lambda^{\Psi} \frac{\partial f}{\partial u} (H_u^{\phi} + \nu^T H_u^{\psi}) \right) dt$$

$$= -k \int_{t_0}^{t_f} \left((H_u^{\phi} + \nu^T H_u^{\psi})^T H_u^{\psi} \right) dt = -k(g + \nu Q)$$

$$\text{Let} \qquad g \equiv \int_{t_0}^{t_f} \left[H_u^{\phi} \right]^T H_u^{\Psi} dt \qquad (q \times 1)$$

$$Q \equiv \int_{t_0}^{t_f} \left[H_u^{\Psi} \right]^T H_u^{\Psi} dt \qquad (q \times q)$$

We want to derive $\delta\psi$ to zero until the procedure is converged.

$$\delta\psi \approx -\eta\psi$$

$$-\eta\psi = -k(g + \nu Q)$$

$$\nu = \left(-\frac{g}{Q} + \frac{\eta\psi}{kQ} \right)$$

$$\delta u = -k\left[H_u^{\phi} + \left(-\frac{g}{Q} + \frac{\eta\psi}{kQ} \right) H_u^{\psi} \right]$$

$$= -k\left[H_u^{\phi} - \frac{g}{Q} H_u^{\psi} \right] - k\frac{\eta\psi}{kQ} H_u^{\psi}$$

$$\therefore \delta u = -k\left[H_u^{\phi} + \nu' H_u^{\psi} \right] - \eta \left(H_u^{\psi} \right)^T Q^{-1}\psi$$

where $\nu' = -\dfrac{g}{Q}$

A Gradient Algorithm (a Direct method)

1) Guess $\vec{u}(t)$
2) Find $\vec{x}(t)$ by forward integration
3) Evaluate ϕ, ψ, $\dfrac{\partial\phi}{\partial x}$, $\dfrac{\partial\psi}{\partial x}$
4) Find H_u^{ϕ}, H_u^{ψ} backward integration.
5) Determine $\nu' = -Q^{-1}g$
6) $\delta u(t) = -k\left[H_u^{\phi}(t) + \nu' H_u^{\psi}(t) \right] - \eta\left[H_u^{\psi}(t) \right]^T Q^{-1}\psi$
7) If $|\delta u(t)| < \varepsilon$, then stop
8) $u(t) = u(t) + \delta u(t)$
9) Go to step 2).

Example:

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = -x_1 + x_1^3 + u, \qquad x_1(0) = 1, \quad x_2(0) = 1$$

$$J = \tfrac{1}{2}\left[x_1^2(1) + x_2^2(1)\right] + \tfrac{1}{2}\int_0^1 \left(x_1^2 + x_2^2 + u^2\right) dt$$

The Hamiltonian

$$H = \tfrac{1}{2}\left(x_1^2 + x_2^2 + u^2\right) + \lambda_1 x_2 + \lambda_2\left(-x_1 + x_1^3 + u\right)$$

The costate equation

$$\dot{\lambda}_1 = -\frac{\partial H}{\partial x_1} = -x_1 + \lambda_2\left(1 - 3x_1^2\right)$$

$$\dot{\lambda}_2 = -\frac{\partial H}{\partial x_2} = -x_2 - \lambda_1$$

$$\lambda_1(1) = \frac{\partial h}{\partial x_1(1)} = x_1(1), \qquad \lambda_2(1) = \frac{\partial h}{\partial x_2(1)} = x_2(1)$$

$$\frac{\partial H}{\partial u} = u + \lambda_2$$

Gradient method for Direct strategy

1) Guess the control $\vec{u}(t)$

2) Integrate the state equation from $t = 0$ to $t = 1$.

   And store the resulting $x_1(t)$, $x_2(t)$

3) Calculate $\lambda_1(1)$, $\lambda_2(1)$ as the substituting $x_1(1)$, $x_2(1)$ from step 2).

   Using $\lambda_1(1)$, $\lambda_2(1)$ as the initial conditions and $x_1(t)$, $x_2(t)$, integrate the costate equations backward on time. And evaluate $\frac{\partial H}{\partial u} = u + \lambda_2$ and store this value.

4) If $\left|\frac{\partial H}{\partial u}\right| \geq \varepsilon$ (tolerance)

   Let $u^{new}(t) = u^{old}(t) - \alpha \frac{\partial H}{\partial u}(t)$

5) Repeat above procedure 2) ~ 4) until $\left|\frac{\partial H}{\partial u}\right| \leq \varepsilon$ (tolerance)

< A Gradient Algorithm for Parameter Optimization with Equality Constraint >

Function minimization

$$Min \quad f(x)$$

with the constraint $L(x) = 0$

$$f(x+\Delta x) \sim f(x) + \frac{\partial f}{\partial x}\Delta x + \frac{1}{2}(\Delta x)^T \frac{\partial^2 f}{\partial x^2}\Delta x$$

$$f(x+\Delta x) - f(x) = \Delta f(x) = f_x \Delta x + \frac{1}{2}(\Delta x)^T f_{xx}\Delta x$$

$$\Rightarrow Min \quad \Delta f(x) = f_x \Delta x + \frac{(\Delta x)^T \Delta x}{2k} \tag{1}$$

subject to $dL = L_x \Delta x = -\eta L$ (2)

when k , η are small constrant.

To solve this linear-quadratic problem, we adjoin the constraint (2) to the performance index with a Lagrange multiplier vector α.

$$\Rightarrow Min \quad \Delta f(x) = f_x \Delta x + \frac{(\Delta x)^T \Delta x}{2k} + \alpha^T(\eta L + L_x \Delta x) \tag{3}$$

For $d(\Delta f) = 0$ with arbitrary $\mathrm{d}(\Delta \mathrm{x})$, it is clearly necessary that

$$f_x + \frac{\Delta x}{k} + \alpha^T L_x = 0 \tag{4}$$

$$\rightarrow \Delta x = -k\left(f_x + \alpha^T L_x\right) \tag{5}$$

Substituting equation (5) into (2), we may solve for α^T

$$L_x\left[-k\left(f_x + \alpha^T L_x\right)\right] = -\eta L$$

$$\alpha^T = \frac{\eta L}{L_x k L_x^T} - \frac{f_x}{L_x}$$

$$= \underbrace{-\frac{f_x}{L_x}}_{\equiv\Lambda} + \frac{\eta L}{k}\frac{1}{\underbrace{L_x L_x^T}_{\equiv Q}}$$

$$= \Lambda + \frac{\eta}{k} L Q^{-1} \tag{6}$$

$$\text{let} \quad \begin{array}{ll} \mathrm{L}_\mathrm{x}^* = \dfrac{1}{\mathrm{L}_\mathrm{x}}, & \Lambda = -\mathrm{f}_\mathrm{x}\mathrm{L}_\mathrm{x}^* \\ \mathrm{Q} = \mathrm{L}_\mathrm{x}\mathrm{L}_\mathrm{x}^\mathrm{T}, & \mathrm{L}_\mathrm{x}^* = \mathrm{L}_\mathrm{x}^\mathrm{T}\mathrm{Q}^{-1} \end{array} \tag{7}$$

Substituting equation (6) and equation (7) into equation (5) gives

$$\begin{aligned} (\Delta x)^T &= -k\left[f_x + \left(\Lambda + \frac{\eta}{k} L Q^{-1} \right) L_x \right] \\ &= -k \underbrace{\left(f_x + \Lambda L_x \right)}_{\equiv H_x} - \eta L \underbrace{Q^{-1} L_x}_{L_x^*} \\ &= -k H_x - \eta L L_x^* \end{aligned} \tag{8}$$

$$\text{where} \quad H \equiv f + \Lambda^T L, \qquad H_x \equiv f_x + \Lambda^T L_x$$

When starting with a possibly poor guess, one can put $k = 0$ and pick $\eta < 1$ to limit the size of Δx, so that the linear approximation is reasonable. Over a few steps, gradually increase η to 1, and a feasible solution ($\mathrm{L} \sim 0$) should be obtained, after which η =1 can be used with $k \neq 0$ to minimize f .

A straight forward numerical method is to guess an initial x and then take small steps Δx in the direction of - H_x^T = the direction of steepest descent. This will lead to a local minimum if one exists. If there are several local minima, this method may not find the global minimum. Choosing the step size requires some experience. If the steps are too large, the minimum may be overshoot, while if the step size are too small, it takes many steps to reach the minimum.

< *Optimization with terminal constraint at open final time* >

(Numerical solution with Gradient method)

Minimize $J = \phi(x(t_f), t_f) + \int_0^{t_f} L(x,u,t)dt$ (1)

$$\dot{x} = f(x,u), \quad x(t_0) = x_0$$

$$\psi(x(t_f), t_f) = 0 \quad t_f \text{ is free}$$

$$J_a = \phi + \nu^T \psi + \int_0^{t_f} \left[L + \Lambda^T (f - \dot{x})\right] dt$$

$$= \phi + \nu^T \psi + \int_0^{t_f} \left[H - \Lambda^T \dot{x}\right] dt$$

$$\text{where} \quad H = L + \Lambda^T f$$

Remember, for t_f is fixed

$$\delta J_a = \delta J_1 + \nu^T \delta\psi$$

$$\text{where} \quad \delta J_1 = \int_{t_0}^{t_f} H_u^{\phi} \delta u dt, \qquad \delta\psi = \int_{t_0}^{t_f} H_u^{\psi} \delta u dt$$

For t_f is free, holding the same notation

$$\Delta J_1 = \dot{\phi} \Delta t_f + \int_{t_0}^{t_f} H_u^{\phi} \Delta u dt + \frac{1}{2k}\left[\int_{t_0}^{t_f} (\Delta u)^T \Delta u dt + (\Delta t_f)^2\right]$$

$$\Delta\psi = \dot{\psi} \Delta t_f + \int_{t_0}^{t_f} H_u^{\psi} \Delta u dt$$

This algorithm may be interpreted as finding a small variation in the control function $\Delta u(t)$, and a small changes in Δt_f to minimize a linear approximation of ΔJ subject to a quadratic penalty on $\Delta u(t)$ and on Δt_f and to a linear approximation to coming closer to the terminal constant $\psi = 0$, namely

$$\min_{\Delta u(t),\Delta t_f} \Delta J \approx \dot{\phi}\,\Delta t_f + \int_{t_0}^{t_f} H_u^{\phi} \Delta u dt + \frac{1}{2k}\left[\int_{t_0}^{t_f} (\Delta u)^T \Delta u dt + (\Delta t_f)^2\right] \tag{3}$$

$$\text{subject to } \Delta\psi = \dot{\psi}\,\Delta t_f + \int_{t_0}^{t_f} H_u^{\psi} \Delta u dt = -\eta\psi \tag{4}$$

where all the coefficient are evaluated at the current estimate of u and t_f and $0 \le \eta \le 1$. k is a scalar step-size parameter. To solve this LQ problem, we adjoin the constraint (4) to the performance index (3) with a Lagrange multiplier vector $\overline{\mathrm{v}}$

$$\Delta\bar{J} \equiv \Delta J + \bar{v}^T\left(\eta\psi + \dot{\psi}\Delta t_f + \int_{t_0}^{t_f} H_u^{\psi} \Delta u dt\right) \tag{5}$$

Now take a differential of $\Delta\bar{J}$

$$\Delta\bar{J} \equiv \Delta\bar{J}(\Delta u, \Delta t_f)$$

$$\begin{aligned} d(\Delta\bar{J}) &= \frac{\partial(\Delta\bar{J})}{\partial(\Delta t_f)} d(\Delta t_f) + \frac{\partial(\Delta\bar{J})}{\partial(\Delta u)} d(\Delta u) \\ &= \left(\dot{\phi} + \bar{v}^T\dot{\psi} + \frac{\Delta t_f}{k}\right) d(\Delta t_f) + \int_{t_0}^{t_f}\left[H_u^{\phi} + \bar{v}H_u^{\psi} + \frac{\Delta u^T}{k}\right] d(\Delta u) dt \qquad (6) \\ &= 0 \end{aligned}$$

For a stationary solution, the coefficient of $d(\Delta t_f)$ and $d(\Delta u)$ must vanish, then,

$$\Delta u = -k\left(H_u^{\phi} + \bar{v}^T H_u^{\psi}\right)^T \tag{7}$$

$$\Delta t_f = -k\left(\dot{\phi} + \bar{v}^T\dot{\psi}\right) \tag{8}$$

Substituting equations (7) and (8) into equation (4) we may solve for $\bar{v}$

$$-k\dot{\psi}\left(\dot{\phi}+\bar{\nu}^T\dot{\psi}\right)-\int_{t_0}^{t_f}\left[H_u^{\psi}k\left(H_u^{\phi}+\bar{\nu}^T H_u^{\psi}\right)\right]dt=-\eta\psi$$

$$k\dot{\psi}\dot{\phi}+k\dot{\psi}\bar{\nu}^T\dot{\psi}+\int_{t_0}^{t_f}\left[kH_u^{\psi}H_u^{\phi}+kH_u^{\psi}\bar{\nu}^T H_u^{\psi}\right]dt=\eta\psi$$

$$k\dot{\psi}\dot{\phi}+k\dot{\psi}\bar{\nu}^T\dot{\psi}+k\int_{t_0}^{t_f}H_u^{\psi}H_u^{\phi}dt+k\bar{\nu}^T\int_{t_0}^{t_f}H_u^{\psi}H_u^{\psi}dt=\eta\psi$$

$$\rightarrow k\bar{\nu}^T\underbrace{\left(\dot{\psi}\dot{\psi}+\int_{t_0}^{t_f}H_u^{\psi}H_u^{\psi}dt\right)}_{Q}+k\underbrace{\left(\dot{\psi}\dot{\phi}+\int_{t_0}^{t_f}H_u^{\psi}H_u^{\phi}dt\right)}_{=g}=\eta\psi$$

$$\rightarrow \bar{\nu}^T=\frac{-kg+\eta\psi}{kQ}=\frac{-g}{Q}+\frac{\eta Q^{-1}\psi}{k}$$

$$=\frac{-g}{Q}+\frac{\eta Q^{-1}\psi}{k}=\nu+\frac{\eta Q^{-1}\psi}{k},\qquad \nu\equiv\frac{-g}{Q} \tag{9}$$

Substituting $\bar{\nu}^T$ into equation (7) and equation (8) gives the desired changes

$$\therefore \Delta u=-k\left(H_u^{\phi}+\nu^T H_u^{\psi}\right)^T-\eta\left(H_u^{\psi}\right)^T Q^{-1}\psi$$

$$\therefore \Delta t_f=-k\left(\dot{\phi}+\nu^T\dot{\psi}\right)-\eta\dot{\psi}^T Q^{-1}\psi$$

Summary

Numerical Gradient Algorithms (Direct method)

1) Choice k and η

2) Guess t_f and $\vec{u}(t)$ $t_0\le t\le t_f$

3) Forward integration $\dot{x}=f(x,u,t)$ and store $x(t)$.

4) Evaluate ϕ, ψ, $\dot{\phi}$, $\dot{\psi}$ at t_f

and determine $\Lambda^{\phi}(t_f)=\phi_x^T(x(t_f),t_f)$

$$\Lambda^{\psi}(t_f)=\psi_x^T(x(t_f),t_f)$$

5) Backward integration for costate equation and store H_u^ϕ, H_u^ψ for $t_0 \le t \le t_f$

$$\dot{\Lambda}^\phi = -L_x - f_x \Lambda^\phi, \qquad \left(H_u^\phi\right)^T = L_u + f_u \Lambda^\phi$$

$$\dot{\Lambda}^\psi = -f_x \Lambda^\phi, \qquad \left(H_u^\psi\right)^T = -f_u \Lambda^\psi$$

6) Determine $\nu = -Q^{-1} g$

where $g = \dot{\psi}\dot{\phi} + \int_{t_0}^{t_f} H_u^\psi \left(H_u^\phi\right)^T dt$

$$Q = \dot{\psi}\dot{\psi}^T + \int_{t_0}^{t_f} H_u^\psi \left(H_u^\psi\right)^T dt$$

7) Compute $\Delta u(t)$ for $t_0 \le t \le t_f$

$$\Delta u = -k\left(H_u^\phi(t) + \nu^T H_u^\psi(t)\right)^T - \eta\left(H_u^\psi(t)\right)^T Q^{-1}\psi$$

8) Compute $\Delta t_f = -k\left(\dot{\phi} + \nu^T \dot{\psi}\right) - \eta \dot{\psi} Q^{-1} \psi$

9) Compute $\delta u_{arg} \equiv \sqrt{\frac{1}{t_f}\int_0^{t_f} \left(\Delta u^T \Delta u\right) dt}$

10) If $\delta u_{arg} < tol_1$ and $\Delta t_f < tol_2$, then stop.

11) Compute new $u(t)$ and t_f

$$u^{new}(t) = u^{old}(t) + \Delta u(t)$$

$$t_f^{new} = t_f^{old} + \Delta t_f$$

12) Go to step 3)